

The Dead Man's Packet

Zone is a phone app that keeps working when every network is gone. Phones talk straight to each other over Bluetooth, and a trapped person's signal **physically walks out of the disaster zone in a rescuer's pocket**. Its one original idea: every packet says when the phone will speak next — so when a phone goes quiet, the network can tell whether that silence was **expected** or whether the phone was **destroyed**.

01

The problem

In the first 72 hours after an earthquake, two failures kill people who were survivable:

- **Communication blackout.** Cell towers are down or overloaded. A person pinned under a slab has a charged phone in their pocket and no way to use it. Rescuers on the pile have no way to hear them.
- **Information fog.** The command post cannot tell **total structural collapse** from **minor waterlogging**, or “**search this whole block**” from “**search this ten-metre circle.**” Crews get sent to buildings that are already clear while calls go unanswered a street away.

Existing mesh-messenger apps (Bridgefy and the like) carry chat between people who are both fine and both holding their phones. They do nothing for someone unconscious, and they treat a phone going offline as a non-event — exactly the event that matters here.

02

What Zone is

One Android APK. Native Kotlin, no servers, no internet — it runs identically in airplane mode. It has three modes, chosen once on first launch:

CITIZEN

The trapped person

Opens straight into broadcasting. Dark screen to save battery and not give away position. One line — “You are being heard” — and one number: how many phones have carried your signal. Three optional buttons: Trapped / Water rising / Safe.

RESPONDER

The rescuer on the pile

Built for gloves, rain and sunlight. A live list of every heard person, sorted by triage priority, not by time. Tap a row for **Dig Here** — a proximity bar that fills as you walk toward the signal. A metal detector, not a map.

MAP / EOC

The command post

A grid-cell severity map for the projector. Cells are coloured by confidence-scored severity. A cell where every phone went silent at once gets its own visual language and a timestamp. A time scrubber replays the whole event.

How a signal leaves the zone

No phone-to-phone pairing, no connection. Every phone continuously broadcasts a tiny packet in its Bluetooth advertisement, and every phone listens. The relay is three moves:

1

Broadcast.

The trapped phone advertises a 31-byte packet — who, where, status, altitude, battery, and when it will transmit again. It repeats on a schedule set by its battery level.

2

Carry.

Any phone in range copies the packet into its store and starts rebroadcasting it with the hop count bumped by one. The original is never modified. Duplicates are dropped by content hash, so the same message never loops.

3

Deliver.

That second phone is walked — by a person, in a pocket — out to a responder or the command post, which now holds the trapped person's packet at hop count 2. The signal crossed the gap on foot.

Merging two phones' stores is just set-union over hash-addressed records — a conflict-free merge (a CRDT) by construction, with no library.

The USP — the Dead Man's Packet

Every packet carries one field the rest of the system is built around: **next_expected_tx** — the number of seconds until the sender intends to transmit again. It is computed honestly from the sender's current battery rung, never faked.

For every phone it has heard, a receiver tracks `last_heard_at`, the promised interval, and last known battery. On a timer it classifies each one:

CONDITION	VERDICT	MEANING
<code>overdue < grace</code>	ALIVE	A missed advert or two. Normal for Bluetooth.
<code>battery ≤ critical</code>	EXPECTED SILENCE	It warned us it was going quiet. Deprioritise, don't alarm.
<code>overdue > 3× promised</code>	UNEXPECTED SILENCE	It promised to speak in 60 s and vanished at 12 with 70% battery. The phone was destroyed. Escalate.

Then the cell-level rule, which is the whole point on the map:

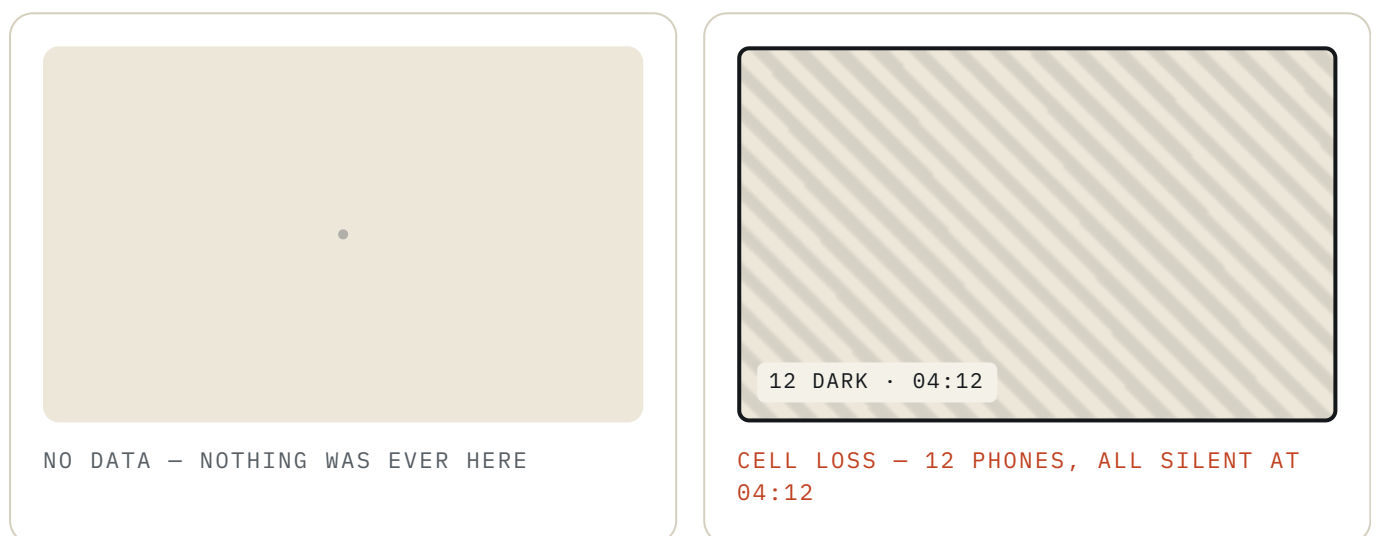
If a ~100 m grid cell holds ≥ 3 phones and $\geq 80\%$ of them enter unexpected silence inside the same 120-second window → **CELL LOSS**. The area collapsed.

Every state change is written to an in-app log with a wall-clock timestamp, so the demo can point at the exact second a phone went dark. This also produces a triage signal no other system has: **you can rank victims by who is about to stop transmitting** — reach the phone at 4% before it goes silent, not after.

05

Silence is the signal

The single most important visual decision in the project: a cell where everyone went quiet at once must never look like a cell that never had coverage. If those two read the same, the idea is invisible on stage.

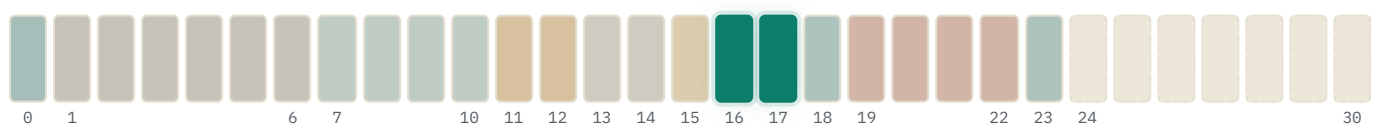


Flat and neutral for absence. A hatched, framed, timestamped hole for a collapse. The map reads at projector distance or it is not doing its job.

06

The 31-byte packet

Bluetooth's legacy advertising payload is 31 bytes. Design for more and you maintain two encoders under pressure, so the core is exactly 31 — “a life in 31 bytes.” Big-endian, carried in the manufacturer-data field.



version / type device id (6 B) position delta (4 B) status · severity / casualties
timestamp battery / hops next_expected_tx — the USP altitude delta + trend
auth (4 B) reserved (7 B)

OFF	FIELD	WHAT IT CARRIES
0	version_type	Protocol version + packet type (0 = status).
1–6	device_id	First 6 bytes of a SHA-256 hash of a per-install random key. The whole identity story — no crypto library.
7–10	position	Lat/lon delta from a local origin, ~1–2 m precision. A reserved value means “no GPS fix” — position is then inferred from the relay path.
11	status	Unknown / Safe / Trapped / Rising water / Injured / Has resource / Responder.
12	severity_casualties	Severity 0–15 and casualty count 0–15 packed into one byte.
13–14	timestamp	Minutes since a fixed event epoch.
15	battery_hops	Battery in 16 steps + hop count 0–15, one byte.
16–17	next_expected_tx	Seconds until the next transmission. The field the whole USP runs on. Must stay honest at every battery level.
18	alt_delta	Relative altitude in metres vs this phone’s launch baseline. A reserved value means “no barometer,” never a false zero.
19–22	auth	First 4 bytes of SHA-256(key payload). Anti-spoofing at demo scale, not real signatures.
23	alt_trend	Metres of altitude change over the last 3 transmissions. Positive = climbing.
24–30	reserved	Zero-filled. Room for a future text-code index.

Content identity hashes over the message bytes but **excludes** the hop nibble and auth, so a message is one record no matter how many times it is relayed.

07

Every feature in the build

TRANSPORT

Connectionless BLE

Broadcast-only advertising in the manufacturer-data field. No GATT, no pairing — far more robust than a connection in rubble.

Dual PHY

Advertises and scans on both 1M and Coded PHY (S=8, eight transmissions per bit for range). Falls back to legacy 1M on chipsets without extended advertising, so the network never silently partitions by handset.

Foreground service

A persistent notification (“You are being heard”) keeps whatever transport is active alive across screen lock — Android kills a background advertiser otherwise.

Transport is an interface

BLE, an in-process 500-node simulator, and a JSON file import/export all implement the same contract. Every piece of logic runs identically on any of them — if Bluetooth fails at 3am, the demo still runs on the simulator.

PACKET & STORE**31-byte codec**

Pure Kotlin, zero Android dependencies, covered by unit tests including a deliberate test that two packets differing only in hop count still hash as one message.

Content-addressed store

Dedup by hash; the same signal heard from five phones is stored once. Merge is set-union — conflict-free by design.

Relay without mutation

Rebroadcasts a hop-incremented copy, round-robin across carried packets so relaying never starves the phone’s own signal. Drops at hop 15.

DEAD MAN'S PACKET**Per-device silence**

ALIVE / OVERDUE / EXPECTED / UNEXPECTED classification on every tick, with a generous grace period tuned on real phones.

Cell-loss detection

The ≥ 3 -phones, $\geq 80\%$, 120-second rule over ~ 100 m grid cells — the collapse signal.

Timestamped transition log

Every state change written to an in-app log with wall-clock time, so the exact moment of going dark is on screen.

TRIAGE

Single-score priority

The responder list sorts by a computed rank, not recency: rising water + climbing altitude first, then just-lost (unexpected silence under 5 min), then trapped below grade, then anyone about to go silent on low battery, then severity.

Reason line

Each row states why it is ranked where it is — “rising water, climbing.”

BAROMETER

Relative altitude

Pressure sensor → metres above or below a baseline captured at launch. Works indoors where GPS is dead. Absolute altitude is discarded — only the delta matters.

Trend & auto-escalation

Altitude change over the last 3 transmissions. A rising trend under “Rising water” escalates to drowning priority with no user input.

Graceful absence

Phones without a barometer are flagged as such, never reported as zero metres.

BATTERY-ADAPTIVE DUTY CYCLE

Interval + PHY ladder

>60% → 1 s on Coded PHY. 30–60% → 10 s alternating. 10–30% → 60 s on 1M. <10% → 300 s, sparse — but still broadcasting an accurate next_expected_tx.

Honest at every level

A phone at 3% that keeps truthfully declaring a 300-second interval is still fully part of the network. A phone that lies here breaks the whole system.

CONFIDENCE & PROXIMITY

Corroboration scoring

Map-cell severity is weighted by how many distinct phones reported it, how many independent relay paths it arrived on, whether the physics is plausible, and whether a verified responder confirmed it. Low-confidence cells render ghosted.

Sybil resistance

Only physically-plausible reports feed the device count, so a flood of packets from fabricated ids can't inflate a cell. A bare "responder" byte earns nothing — trust comes only from a responder packet that verifies against the shared key.

Near-damage alert

A phone that finds itself one hop from a collapsed cell tightens its own beacon interval — the network densifies exactly where it matters.

THREE INTERFACES**Citizen**

One headline, one number, three buttons. The count springs and the phone buzzes the instant another phone picks up the signal.

Responder

High-contrast daylight list; rows slide to their new rank as triage changes. Dig Here proximity bar with a link-quality readout and a haptic tick at each range band.

Map

Severity grid, distinct silence rendering, coverage-gap hints, a "send here first" priority strip, and a replay scrubber.

DEMO & SCALE**Seeded simulator**

500 synthetic nodes over Rasuwagadhi geography, replaying the real toll curve (22 → 95 → 469 → 626 over 72 hours) compressed to about 90 seconds, with a scripted mass-silence event so cell loss fires on cue.

Browser EOC viewer

A zero-dependency PWA that draws the severity map and triage list from a snapshot. Opened as `?live=eoc.json` it polls a same-origin file every 3 s — the phone rewrites that snapshot every 4 s, so a laptop at the command post tracks the field with no network.

Hidden battery override

A long-press gesture cycles a faked battery level so the duty-cycle ladder can be shown on stage on fully charged phones.

SINCE THE HACKATHON BUILD (V0.3)**Crash-safe state**

The store, the confirmed collapses and the silence tracks are written to disk and reloaded on start, so a reboot or an OS kill mid-incident doesn't wipe a carrier phone. Atomic write, 24 h age cutoff.

Real position	The heartbeat sends an actual GPS fix (last-known, no Play Services, no network) as a delta from a configurable per-incident origin — falling back to no-fix, where the relay path localises the device.
Inferred TRAPPED	A phone held dead still for five minutes, with no user input, broadcasts TRAPPED_DEBRIS — the “works when unconscious” claim, made true.
Pre-shared responder key	A responder phone provisioned with a shared secret signs its packets with a SHA-256 MAC any phone can verify (no crypto library). A forged “responder” is rejected; a real one lifts a cell’s confidence.
RESOLVE	A responder marks a signal reached — a signed type-1 packet carrying the target’s content-id. It rides the same store-carry-forward, and every phone and the EOC drop that report, so the picture converges on who still needs help.
Hot / cold Dig Here	A least-squares fit over the RSSI slope while the responder walks — WARMER / HOLDING / COLDER — so the last few steps read as toward the signal or away from it.
Headcount	An optional “people with you” stepper on the Citizen screen feeds the packet’s casualty count.
Sneakernet	Export the whole mesh state to a JSON file, share it by any means the phone has, import and set-union merge it on another phone — the fallback when BLE is dead and there is no cable.
Language	The Citizen screen — the one a local victim reads — in English, Hindi and Nepali, picked on the landing screen.
Field readiness	Real app identity and adaptive icon; an on-device crash log (no analytics, no network); a “Keep Zone alive” screen that walks the Doze exemption and per-OEM autostart settings; CI running the unit tests on every push.

08

What we deliberately left out

Stated up front, because a limitation you declare is respected and one that is discovered is not.

CUT	WHY IT IS DEFENSIBLE
iOS devices	iOS Safari has no Web Bluetooth at all, and native iOS background advertising is locked down by Apple. Two iOS phones would both be listeners and never hear each other. A real deployment needs an Apple entitlement or a hardware beacon.
Per-device signatures	The RESPONDER tier now verifies against a pre-shared key (a SHA-256 MAC — no library). Full per-device Ed25519 identity is still cut: it needs a crypto dependency and more than 31 bytes.
RSSI trilateration	A proximity bar plus a hot/cold trend converts “this block” into “this 10 m circle.” True trilateration needs movement tracking that would not have been finished.
A CRDT library	Merging hash-addressed records by union already is a CRDT. Importing one would add weight and nothing else.
An internet gateway	Out of scope by design. The responder’s own phone showing “47 signals in this building, 3 rising-water” is already the whole product. A continuous local snapshot and a sneakernet file cover the command-post hop; a bearer to the wider world is future work.
Rubble penetration claims	2.4 GHz attenuates hard through concrete and rebar. The answer is not magic range — it is that a pinned phone advertises for days at low power, and a rescuer walking the pile eventually gets within 10 m.

09

Proven on real hardware

Validated on three phones — Samsung Galaxy S23 (trapped), Realme 7 (carrier), Motorola Edge 50 Pro (responder) — in airplane mode, on 2026-08-30.

TEST	RESULT	EVIDENCE
Direct link	PASS	Motorola heard both phones at 1 hop, marked LIVE.
Coded PHY reception	PASS	Both PHYs confirmed received on the Motorola.
Store-carry-forward — the demo	PASS	S23 with Bluetooth off; its Trapped signal still reached the Motorola at 2 hops, relayed by the Realme.
Unexpected silence	PASS	S23 at healthy battery, powered off → row went red SILENT with the wall-clock time it went dark.

TEST	RESULT	EVIDENCE
Expected silence	PASS	S23 at 8% (declares 300 s), powered off → row went grey EXPECTED, no escalation. The two are told apart.
Barometer	PASS · borderline	One floor read +2 m (spec is ~±1 m). Sign and magnitude correct; basement/rooftop separation solid.
Drowning escalation	PASS	“Water rising” + climbing → S23 jumped to the top of the list.
Dig Here	PASS	Proximity bar tracks walking toward and away.
Cell loss	PASS	Verified in the 500-node simulator — one cell flips to a hatched, framed “12 dark” hole.

The v0.3 hardening (persistence, real GPS, inferred TRAPPED, responder key, RESOLVE, hot/cold Dig Here, live EOC, sneakernet, Hindi/Nepali) was re-checked on the Realme 7 on 2026-09-04: real GPS fix read and encoded, the battery ladder picking the right interval and PHY off the actual charge, the responder key verifying, no crashes. The mesh half — 2-hop relay, RESOLVE propagation, a verified responder lifting another phone’s map — re-runs on the full three-phone set. Also outstanding: a 30-minute soak run and a range survey.

One binary, three roles. No servers, no internet, no analytics, no downloaded map tiles. It works in airplane mode or it does not ship.

ZONE · NATIVE KOTLIN + JETPACK COMPOSE · minSdk 26 · BLE store-carry-forward · 31-byte packet contract